

The AI Revolution in Data Storytelling

We are all—both you and I—living through a revolution. And true revolutions have been rare in history. Somewhere, I once heard that there have been only three genuine revolutions: the invention of writing about 5,000 years ago, which marked the shift from prehistory to history; the Industrial Revolution, roughly 300 years ago, which moved us from manual labor to mechanization; and today, the rise of generative Artificial Intelligence (AI).

To be precise, rather than mere revolutions, these are epochal transitions—moments when one era ends and another begins. The invention of writing changed everything. Apart from archaeological findings, we know little about how people lived before writing. But with writing, humanity began to record thoughts, ideas, and knowledge. It's unlikely that early humans spent time contemplating grand philosophical concepts—or at least if they did, we have no record of it because they couldn't write it down.

The Industrial Revolution marked another massive shift as machines extended human muscle power, enabling large-scale production. Now, with generative AI, we are witnessing yet another fundamental transformation: a powerful "macro brain" is augmenting our individual minds, helping us think, create, and process vast amounts of information. Generative AI is, in a way, the result of humanity's relentless pursuit of knowledge—it holds billions of pieces of information that no single brain could ever retain. It's like one of those colossal mining machines that can lift the weight of a

thousand men, but instead of lifting material, it processes and synthesizes human thought on an unimaginable scale. Since we are in the midst of this profound shift, we cannot afford to ignore it. The risk is being left behind, clinging to the past while the world moves forward. Instead, we must embrace this transformation—ride the wave, even if the sea is wild and unpredictable.

If you are reading this chapter, you are undoubtedly interested in understanding how to make the most of generative AI in data storytelling. This is exactly what you will see now. In particular, in this chapter, you will see:

- Different ways of using AI in data storytelling
- How AI enhances Data Storytelling
- Using AI for story assessment
- Ethical considerations in the use of AI in Data Storytelling

Let's start from the first point, the different ways of using AI in data storytelling.

Different Ways to Use AI in Data Storytelling

You can use AI in data storytelling in four ways (Li et al., 2024), as shown in Figure 13.1.

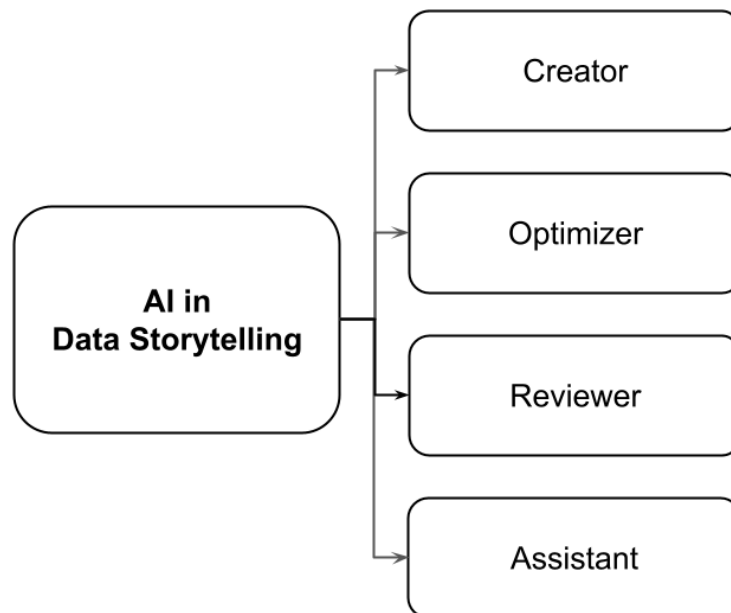


Figure 13.1 The different ways of combining AI and data storytelling.

AI, when used as a *creator*, is responsible for generating narrative or visual content. This approach is handy for quickly producing draft articles, slides, or infographics, starting from structured inputs or datasets. One of the main advantages of this modality

is time-saving: tools like ChatGPT can write a textual description of a dataset in a few seconds, while models like DALL-E can generate images without the need for a human designer. However, the quality of these outputs is not always satisfactory and often requires corrective interventions. Furthermore, the risk of leaving everything to AI is the production of flat and unoriginal content. For example, on the web, you see many similar texts generated with AI that no longer affect the reader. This is because if you don't guide the AI to create the text or image you want, the AI uses its *default style*, which is always the same. Use AI as a content creator to get inspiration or speed up the process of creating drafts, but not final documents.

A second way to use AI is as an *optimizer*. In this case, AI does not generate content from scratch but intervenes to refine the material you have produced. Use this technique to improve the readability of texts, refine visualizations, and suggest stylistic changes. Personally, I find this mode very useful when I have to explain concepts: in fact, I ask the AI to rewrite them so that they are more understandable. Also, this mode can be used to match the tone of a presentation to the type of audience. The biggest problem in this case is the AI's difficulty in understanding the author's context and intentions. This means automatic optimization is helpful but must be accompanied by careful supervision.

The third way of using AI is as a *reviewer* of the content you create so you can identify potential problems. This may include checking for incorrect data, identifying narrative inconsistencies, or proposing alternatives to improve clarity. Therefore, this role of AI is particularly effective as a support tool but cannot replace your critical analysis.

The last role is that of *assistant*, in which AI does not make autonomous decisions but provides support in specific tasks, such as collecting data, summarizing documents, drafting texts, generating interactive visualizations, or carrying out repetitive and low-value-added tasks, such as formatting graphs or translating texts for different markets. This role values AI as an intelligent automation tool, reducing your workload.

The analysis conducted suggests that, in data storytelling, AI is more effective when it operates in collaboration with humans rather than replacing them. The four modalities identified – creator, optimizer, reviewer, and assistant – each offer advantages and limitations, and the success of their use depends on the ability to integrate these tools into a broader process of human creation and review. In this chapter, we will use AI as an assistant because it will help us create and adapt the story to the audience. Let's see how.

How AI Enhances Data Storytelling

When you create a story, you first design it (making phase) and then adapt it to your audience (delivering). The making phase is divided into pre-production, production, and

post-production. In preproduction, you define the theme, which includes the central idea extracted from the data, the message you want to convey, and the values that guide the story. Next, you develop the plot, identifying the protagonist (the hero), their goal, and the obstacles they must overcome. Finally, write the script detailing the scenes following the Three-Acts structure: introduction, conflict, and resolution.

Use AI in the design phase to define characters and plot. AI can generate plot drafts following the Three-Acts structure, suggesting how to evolve the story from a narrative point of view. It can also propose story variations based on different scenarios or simulations of the data. Later in the chapter, you will see practical examples of how to use AI in this case.

During the delivering phase, you refine the story based on your audience. Audience analysis is, therefore, fundamental in this phase to understand precisely who you are in front of. Use AI in the delivery phase to adapt the story to your audience, specifying the context, tone, and language to use. AI can analyze the target audience, suggesting changes to the language to make it more accessible or technical as needed. For example, the AI might recommend more detailed and technical language for an expert audience while suggesting a more straightforward tone for a general audience. Furthermore, AI can personalize the presentation based on communication channels (email, presentations, videos), optimizing the format and visual structure of the contents. Later in the chapter, you will see a practical example of using AI to adapt the story to the audience.

Let's now see how to use AI in the two phases, starting from the first phase, the making phase.

Using AI during the Making Phase

During this phase, first, you define the characters and the plot. Since character definition is very delicate, I suggest you do it by hand, without using AI. However, if you feel inspired, you can ask the AI to extract characters from the data. But this is at your own risk. In this section, however, you will see how to use AI to speed up the creation of the plot, using the Three-Acts structure, assuming that you already have a clear idea of who the main characters are (at least the hero and the antagonist). To understand how to proceed practically, I will use the example of the story of temperature increase. In the example, I will use ChatGPT in the Plus version, but you can also repeat the same experiment with other generative AI tools, such as Gemini, Claude, etc. Suppose you have data on rising temperatures. In this case, the hero is the temperature itself, while the antagonist, for example, is CO₂ emissions. The problem that the hero encounters is the risky increase, which can lead to problems for the whole planet. Once you've defined the hero, the antagonist, and the problem, use AI to structure the plot into Three Acts. Proceed as follows.

First, open ChatGPT and start a new conversation by typing the following prompt:

I want to create a story about climate change using the three-act structure. The story's hero is temperature, and the antagonist is CO₂ emissions. The problem the hero faces is that the temperature is increasing. Can you help me define the plot? Don't humanize characters, but describe facts from a scientific perspective.

With this prompt, ask ChatGPT to create a story based on the Three-Acts structure, with temperature as the hero, CO₂ emissions as the antagonist, and climate change as the topic. The prompt also specifies the problem that the hero has and, *possibly*, the solution. I say *possibly* because you can also not specify the solution to see what ChatGPT proposes as a solution. In my case, in fact, I did not include the solution. Then, you will specify further details. However, the more information you specify at this stage, the more detailed the plot will be produced.

ChatGPT responds by generating a story structured in three acts since it knows this technique from cinema and novels. In my case, ChatGPT generated the story described in Table 13.1.

Table 4.1: The story generated by ChatGPT

ACT	STRUCTURE
First Act The Stable Balance (Setup)	Introduction to the World: The Earth's climate system has maintained a relatively stable equilibrium for thousands of years. Temperature fluctuates within natural cycles (like glacial and interglacial periods), influenced by factors such as solar radiation, volcanic activity, and natural greenhouse gases. This balance allows for diverse ecosystems and stable sea levels, supporting life as we know it. Introduction of the Antagonist: During the Industrial Revolution, CO ₂ Emissions begin to rise sharply due to human activities like burning fossil fuels (coal, oil, and natural gas), deforestation, and industrial processes. Initially, the impact is subtle, with only slight increases in atmospheric CO ₂ levels. Inciting Incident: By the mid-20th century, atmospheric CO ₂ concentrations surpass 300 ppm, levels unseen for millions of years. This marks the start of a disruption in Earth's energy balance. Temperature starts to respond, warming slightly but steadily, signaling the beginning of a broader climatic shift.
Second Act The Warming Conflict (Confrontation)	Rising Action: CO ₂ Emissions continue to accumulate, enhancing the greenhouse effect. This traps more heat in Earth's atmosphere, causing Temperature to rise at an unprecedented rate. Glaciers and ice sheets begin to melt, contributing to sea-level rise. Ocean temperatures increase, leading to coral bleaching and disrupted marine ecosystems. Heatwaves, wildfires, and extreme weather events become more frequent and severe. The Tipping Points: Temperature approaches critical thresholds. The melting of permafrost releases methane—a potent greenhouse gas—further accelerating warming in a dangerous

feedback loop. The loss of Arctic sea ice reduces Earth's albedo (reflectivity), causing even more heat absorption. CO2 Emissions, emboldened by continued fossil fuel consumption, push the climate system toward potential points of no return.

Midpoint Crisis:

Despite growing scientific evidence and international efforts like the Kyoto Protocol and the Paris Agreement, global emissions continue to rise. Temperature crosses the 1°C mark above pre-industrial levels. The impacts become undeniable: stronger hurricanes, prolonged droughts, and massive biodiversity loss. Ecosystems struggle to adapt, and vulnerable regions face existential threats.

Third Act
The Heat of
Resolution
(Resolution)**Climax:**

The world reaches a pivotal moment. Atmospheric CO2 surpasses 400 ppm. Temperature edges dangerously close to 1.5°C above pre-industrial levels—the limit scientists warn must not be crossed to avoid catastrophic impacts. However, a surge in renewable energy adoption, reforestation efforts, and emission reduction policies begin to slow the rate of increase. The global community starts to take serious, coordinated action.

Falling Action:

The growth of CO2 Emissions stabilizes and begins to decline due to aggressive mitigation strategies: carbon capture technology, transition to clean energy, and widespread conservation efforts. Temperature continues to rise for a while due to the inertia of the climate system but begins to plateau. The damage is significant but not irreversible in all areas.

Resolution:

Temperature stabilizes below 2°C of warming, avoiding the worst-case scenarios, though many ecosystems and communities have been forever changed. The story ends with a cautious sense of hope, emphasizing the resilience of natural systems when given the chance to recover. The battle against CO2 Emissions isn't over, but the path toward a more stable climate has been set.

As you can see from Table 13.1, ChatGPT has generated a story framework, which still needs revision but can serve as a starting point. The structure proposed by ChatGPT is slightly different from the one I defined, but it is still valid. For example, there are no pinch points explicitly.

At this point, you can ask ChatGPT to add more characters to your story or edit pieces of the story. For example, if you want to add a sidekick, write this prompt still in the same conversation where ChatGPT generated the plot:

Add a sidekick to the story. The sidekick should be real and not a fictional character.

In my case, ChatGPT added the following scene in Act Two after the Rising Action:

The Sidekick Steps Up:

Countries begin investing in Renewable Energy, recognizing its potential to combat CO2 Emissions. Wind farms sprout across landscapes, solar panels cover rooftops, and innovations in battery storage improve reliability. Despite this, fossil fuel industries resist, and global emissions continue to rise, creating tension between progress and inertia.

As a sidekick, ChatGPT has added countries that are starting to invest in renewable energy. If you don't like this sidekick or if it doesn't help you in your story, you can always ask ChatGPT to edit it and add the sidekick you prefer. Always remember that when you use ChatGPT to generate stories, you risk obtaining fictitious stories that are not anchored to the data, so I recommend you always verify the story produced. For example, you can ask ChatGPT to replace the proposed sidekick with a scientist's testimony you previously collected. Use the following prompt:

Replace your sidekick with a scientist I have interviewed who has listed the problems relating to temperature increase.

In my case, ChatGPT removed the previous sidekick and added a new sidekick to the First Act immediately before the inciting incident. In particular, it generated this piece of history:

The Scientist Appears:

Enter the Scientist, a researcher dedicated to understanding Earth's climate systems. Early in their career, they observe minor anomalies in global temperatures and atmospheric CO2 levels. Their initial studies reveal troubling trends, but widespread attention remains elusive.

If you still want to refine the story, you can continue your dialogue with ChatGPT and ask it to modify or better specify some parts. The idea is that you always lead the generation of the story. ChatGPT is simply your assistant, helping you make the story more compelling. This phase aims to use AI to define the plot and add parts to your story. AI is a great servant and a terrible master, so use it wisely.

In summary, to integrate AI into the making phase, follow the steps specified in Figure 13.2.

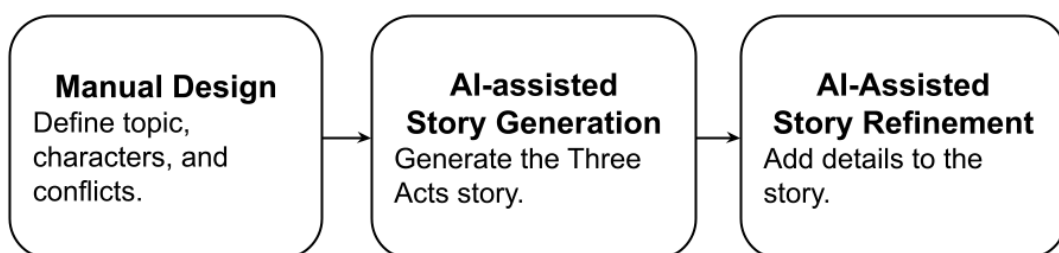


Figure 13.2 The steps to follow to integrate AI into the making phase.

First, extract the main characters from the data - the hero and the antagonist. Also, define the problem the hero has and the possible solution. Do this manually, without using AI, to avoid extracting fictitious characters without any basis in the data. Also, define the story's topic (for example, climate change), possibly specifying details on how your story should unfold. Then, ask the AI to generate the story for you, dividing it

into three acts. You can also specify the specific structure you want the AI to generate, such as the presence of a hook, plot points, etc. Add details and additional characters, or edit parts that don't align with your story until you achieve the desired result.

Once the plot and characters are defined, you can move on to the production phase, where you implement the individual scenes of the story. AI can help you here, too, but this depends on the specific tool you use to implement the scenes. For example, in my book, *Data Storytelling with Altair and AI* (Lo Duca, 2024), I explain how to use AI to implement scenes using Python. Once you have implemented all the scenes, you can move on to the post-production phase, where you proceed to edit the story. Depending on your chosen tool, you can use AI as an assistant or proceed manually. After editing, you can finally refine your story for the audience, moving on to the delivery phase. Let's now see how you can use AI during this phase.

Using AI during the Delivering Phase

In the delivering phase, you adapt your story to the audience. AI can help you understand who your audience is and refine the previously generated storyline for your audience. Suppose you want to adapt the story about rising temperatures for an uninterested general audience. Ask the AI to insert a hook at the beginning of the story to capture the attention of this type of audience. Write the following prompt, continuing the conversation with ChatGPT:

List 10 possible hooks for a generic audience disinterested in climate change.

In my case, ChatGPT provided the following hooks organized by type:

1. The Extreme Weather Hook

"Remember that scorching heatwave last summer? Or that unexpected snowstorm? What if I told you these are just the beginning of a much bigger shift?"

2. The Money Hook

"Climate change isn't just about polar bears—it's about your wallet. Rising temperatures mean higher electricity bills, pricier groceries, and costly natural disasters. Ready to find out why?"

3. The Food & Drink Hook

"Your morning coffee, your favorite chocolate, even your weekend beer—climate change is making them harder to produce and more expensive. Here's why that matters to you."

4. The Health Hook

"What if the air you breathe and the water you drink could make you sick? Rising temperatures are fueling pollution, disease, and heat-related illnesses. Here's what's at stake."

5. *The Sports Hook*

"The future of football, skiing, and even the Olympics is at risk. Rising temperatures and extreme weather are already reshaping the sports world—here's how."

6. *The Travel Hook*

"Dreaming of a beach vacation or a ski trip? Many of the world's most beautiful destinations—from tropical islands to snowy peaks—are disappearing due to climate change. See what's vanishing."

7. *The Tech Hook*

"From electric cars to solar-powered gadgets, climate change is driving an innovation boom. The future is here—are you ready for it?"

8. *The History Hook*

"The last time Earth's temperature rose this fast, dinosaurs went extinct. What's different now? Humans are the cause—and we still have time to change the outcome."

9. *The Space Hook*

"Mars and Venus are both lifeless planets—but for different reasons. Earth is heading toward a dangerous crossroads. Which path will we take?"

10. *The Mystery Hook*

"There's an invisible force changing our planet every second. It's altering the weather, reshaping oceans, and affecting our lives in ways we don't even notice—until it's too late. What is it?"

If you like a particular hook, ask ChatGPT to refine it. For example, use the following prompt to structure the hook at the beginning of an oral presentation:

Structure Hook 2 for one minute of a longer oral presentation about the story of climate change.

In my case, ChatGPT provided an interesting answer:

Opening Line (Attention Grabber):

"What if I told you that climate change is silently draining your wallet? You don't have to care about the environment to feel its impact—because it's already costing you more every day."

Key Points (Concise & Engaging):

Higher Energy Bills:

"Heatwaves make summers hotter, winters are becoming more unpredictable, and that means you're paying more to keep your home cool and warm. In some areas, energy prices have spiked because of extreme weather disrupting supply chains."

Rising Grocery Prices:

"Droughts, floods, and wildfires are destroying crops. That means your morning coffee, your favorite snacks, even basic groceries—are getting more expensive. In the last decade alone, food prices have risen in part due to climate disruptions."

Costly Climate Disasters:

"More frequent hurricanes, floods, and wildfires aren't just tragic—they're expensive. Insurance companies are raising rates or pulling out of high-risk areas. Homeowners and businesses are left paying the price."

Transition into the Main Presentation:

"Climate change isn't just a problem for future generations—it's hitting us right now. But why is this happening? To understand that, we need to look at the story of climate change: how temperature—the hero of our story—is battling against an invisible force that threatens to change the world as we know it."

You can modify the organization proposed by the AI directly by entering your changes or asking the AI itself to make the changes for you.

In addition to refining the hook based on the audience, AI can also adapt the context and other parts of the story to the audience to make it familiar to that type of audience. You can ask the AI to provide details on the habits and customs of a particular kind of audience. For example, if you are dealing with an audience from a specific country, you can ask for culinary information, customs, idioms, etc. Then, you can ask the AI to generate examples based on these audience-specific traditions integrated into the plot of your story. You will see that the results will be shocking.

A further step is to use AI to calibrate the tone and language of the story. You can ask AI to transform your story from informative to persuasive, achieving shocking results. You can ask AI to translate it into another language to turn it into a funny or extremely sad story. AI is at your service. In summary, in the delivery phase, use AI to adapt your story to the audience, such as the hook, context, or tone throughout the story.

I haven't told you so far that AI is beneficial not only for story construction and refinement but also for evaluating the quality of the story produced. Let's see how in the next section.

Using AI for Story Assessment

In 2024, I attended the Discovery Science conference (2024) where I had to present a poster on integrating AI and environmental data for automatic report generation. Among the various speakers, a professor presented a paper in which she used AI to classify some reviews and then used the same AI to evaluate the performances obtained (Cosentino et al., 2024). This intrigued me a lot, and then I thought that using AI to assess something could also be applied to data storytelling. And then, I developed the idea I will explain to you.

Imagine having a story ready and adapted for a specific type of audience. Before administering your story to the audience, it would be ideal to test its effectiveness. A

completely unexplored terrain in literature is using AI to evaluate the quality of a story for your audience. Follow the steps described in Figure 13.3.

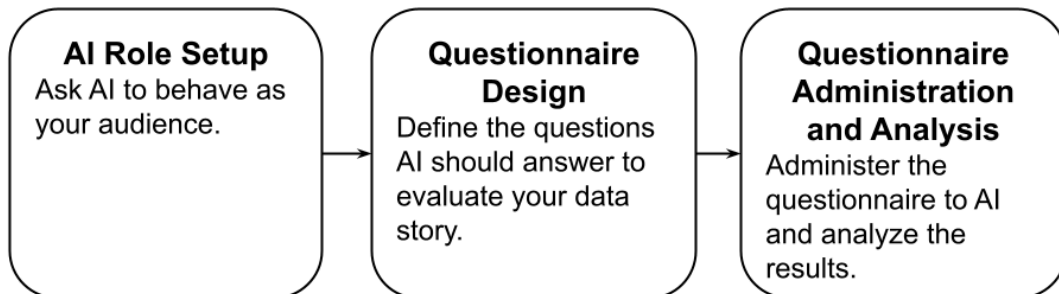


Figure 13.3 The steps to follow to integrate AI into the evaluation phase of a story.

Suppose you already have a clear idea of who your audience will be. First, ask the AI to take on the role of your audience. The more details you specify, the higher the level of personification you will achieve. Next, prepare a questionnaire containing a series of questions related to your story to provide to the simulated audience, such as whether they understood the story's context, intended to take the suggested next steps, etc. Then, administer the questionnaire and story to the AI and ask the AI to answer your questions, perhaps assigning a score to each answer on a scale from 1 (not very likely) to 5 (very likely). Finally, analyze the results generated by AI. These can give you an idea of whether your story is relevant to that audience. Of course, the AI might make mistakes, but in any case, it can give you an idea of the story's weak points.

Let's now look more at an example of a questionnaire you can administer to the AI. Table 4.2 shows a possible questionnaire divided by topic. Each topic has multiple questions associated with it.

Table 4.2: An example of questions to administer to AI

TOPIC	QUESTIONS
Understanding the Story	Was the central insight of the data story clear and understandable? (Scale 1-5) What do you think was the key takeaway from the data? Were there any parts of the data story that seemed unclear or difficult to understand? If so, which ones? Did you understand the context and background of the data without difficulty? (Scale 1-5)
Engagement and Interest	Did the data story capture your attention from the beginning? (Scale 1-5) Which part of the story was most engaging for you, and why? Did the visualizations and narrative keep you interested throughout? (Scale 1-5) Did you need to stop reading or disengage before concluding? (Scale 1-5) Did the story make you curious to explore the topic more?
Emotional and Cognitive Impact	What emotions did the data story evoke (e.g., surprise, concern, curiosity)? Were there specific insights that made a strong impression on you? If so, which ones? How engaging was the data story on an emotional or intellectual level? (Scale 1-5) Did the data story challenge your previous assumptions or perspectives? (Scale 1-5)

Relevance to the Audience	Was the data story relevant to someone with your interests or background? (Scale 1-5) Did you feel that the examples, comparisons, or case studies used were meaningful? (Scale 1-5) Was the tone and style appropriate for the target audience? (Scale 1-5) Which elements of the data story felt most relevant, and which felt less so?
Message and Call to Action	What was the main message or conclusion of the data story? Did it encourage you to think differently about the topic? If so, how? Did the story inspire you to take action or decide based on the insights presented? (Scale 1-5) If a specific action was suggested (e.g., policy change, behavior shift, further research), how likely are you to follow it? (Scale 1-5) After reading the data story, do you feel more informed, persuaded, or motivated?
Quality of Narrative and Visualization	Was the pacing of the data story appropriate? (Scale 1-5) Were the data visualizations clear, effective, and easy to interpret? (Scale 1-5) Did the combination of text, visuals, and numbers enhance understanding? (Scale 1-5) Were any parts of the story too technical or difficult to follow? If so, which ones? Did the story maintain a balance between data accuracy and compelling storytelling?
Overall Assessment	How would you rate the overall quality of this data story? (Scale 1-5) Would you recommend this data story to someone interested in the topic? Why or why not? What improvements would make the story more compelling or engaging?

The first question to ask is: is the main message clear? Can the public understand the context and meaning of the data without having to do mental acrobatics? If the answer is negative, it could indicate that the data is too complex or that the narrative is poorly structured. Ask the following key questions: *What is the main message you took away from the story? Was there anything that seemed confusing or unclear to you?* If the answers are vague or inconsistent, it's time to revise your story to make it clearer and more impactful.

The second group of questions focuses on engagement and interest. Effective data storytelling is not only informative but also captures attention and holds it until the end. Why? Because data needs to be told like a great story, with tension, curiosity, and surprising revelations. If the audience loses interest halfway through, the problem could be an introduction that is too long, an ineffective visualization, or an unexciting use of data.

The third group of questions concerns the emotional and cognitive impact. The numbers, in themselves, do not excite. But a good story with data can surprise, worry, and make you think. When we read that 90% of people don't use their data to make strategic decisions, do we feel part of the problem? Or do we remain indifferent?

Next, you can investigate the relevance to the audience. A compelling story must be relevant. Even the most engaging visualization will be useless if the audience does not perceive it as concerning them. If the audience responds that they don't feel involved, you may need to redefine better who you are telling this story for. Then, you analyze the final message and the call to action. Good data storytelling is not an end in itself; it also serves to push for action. After reading your story, is the audience motivated to do

something concrete? If the audience says “nice, interesting” and then moves on to something else, something hasn't worked. If the responses are lukewarm, the story may need a more decisive close or explicit message.

As the penultimate topic, analyze the quality of narrative and visualization. Data stories are made of words, numbers, and visualizations. But be careful: a poorly designed graphic can destroy the message. If the visualizations are poorly designed, the entire narrative can lose effectiveness.

Finally comes the crucial question: how effective is this story? It's time to take stock with an overall evaluation. If the average score is low, there is room for improvement. And that's great news: It means you have the power to refine your storytelling to make it even more effective. Testing a story with these questions is like testing it before the official launch. The collected answers can reveal hidden weaknesses, suggest improvements, and, above all, ensure that the public understands the data and does something concrete with it. In the end, the real goal of data storytelling isn't just telling the numbers. It's about changing the way people see the world.

Suppose you want to administer the questionnaire to the head of a company that produces shoes. Suppose you implemented your story as a presentation and wrote the questionnaire in a text file. Write the following prompt in ChatGPT plus, attaching the two documents (presentation and questionnaire):

Suppose you are the head of a company that produces shoes, and you have to decide about the next investments. Consider the attached presentation and answer the attached questions.

ChatGPT will provide you with the answer to the questions. Analyzing the responses lets you understand if your story is clear to your audience.

At this point, you may have asked yourself: can I use AI in data storytelling? In other words, are the results produced by AI reliable, or could they present some problems? Let's find out in the next section.

Ethical Considerations in Using AI in Data Storytelling

AI in data storytelling can speed up the creation of stories. However, integrating AI into this space is not without risks. While on the one hand, AI can enhance the ability to describe data clearly and effectively, on the other, it can introduce problems related to toxicity, incompleteness, inaccuracy, and bias, influencing the perception of reality and the quality of information. To ensure AI's ethical and responsible use in data storytelling, measure the four key metrics described in Figure 13.4.

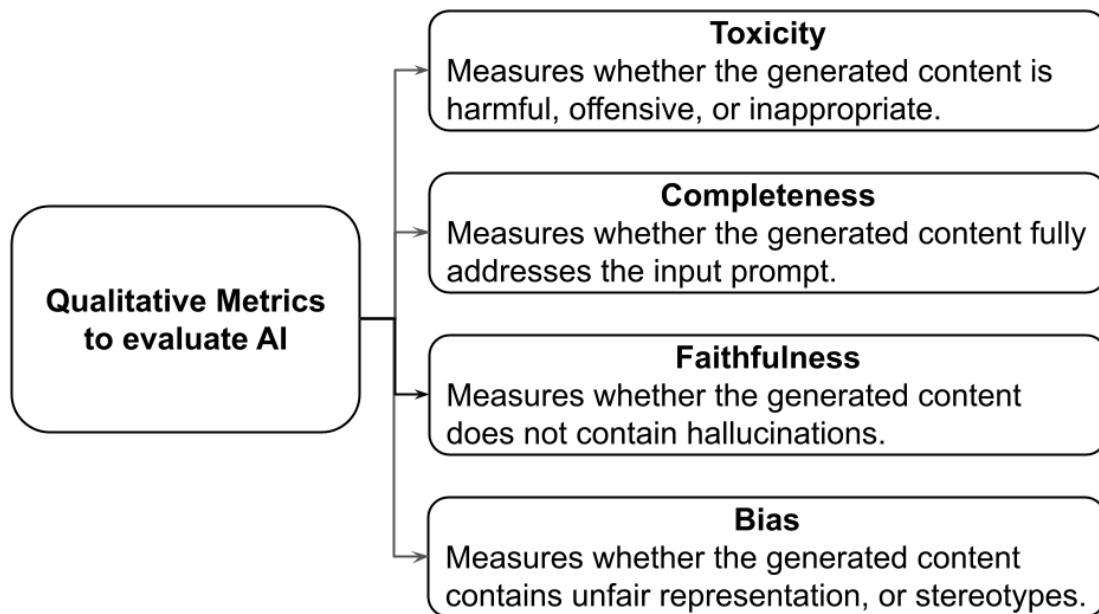


Figure 13.4 The four metrics to evaluate AI-generated content.

The *toxicity* measures whether AI-generated content is harmful, offensive, or inappropriate. The risk of toxicity is particularly relevant when dealing with sensitive issues, such as climate change, public health, or social inequalities. To reduce this risk, manually review the generated content to ensure the message is appropriate and does not exploit the data. Respectful and neutral storytelling improves credibility and reduces the risk of negatively influencing the audience.

The second aspect concerns the *completeness* of the generated output, which evaluates whether it fully responds to the input prompt. To assess the completeness of an AI-generated story, check the content produced with the original request and check whether all relevant information has been included. You can also use automatic scoring tools, which measure the percentage of key points covered.

Faithfulness is the third aspect to analyze, which measures whether AI-generated content contains hallucinations, producing inaccurate or completely invented content. This aspect is particularly critical in data storytelling since the story must be based exclusively on verified data. To minimize the risk of incorrect information, compare the generated content with reliable sources and adopt approaches based on Retrieval-Augmented Generation (RAG) (Lewis et al., 2020), which allows AI to anchor its responses to actual data. If a data story makes claims unsupported by data, the risk of misinformation increases and the audience could be misled into making decisions based on incorrect information.

The last aspect to consider is *bias*, that is, the presence of unfair representations or stereotypes. AI, being trained on human data, inevitably reflects some pre-existing

biases. This can lead to the creation of stories that emphasize certain narratives at the expense of others, introducing distortions into the representation of reality. To identify and correct bias in AI-generated stories, constantly review the generated content using bias detection models, which analyze language and content for imbalances. Another effective strategy is demographic parity analysis, which allows you to check whether all social groups are equally represented in the narrative. Bias-free data storytelling guarantees a more fair and inclusive representation of the phenomena described, avoiding reinforcing stereotypes or systemic injustices.

In summary, to ensure the responsible use of AI in creating data-driven stories, take an approach that combines automation and human supervision. Follow the steps described below:

- Review any AI-generated content for quality and accuracy.
- Use AI as a support tool, not as a substitute for your critical thinking.
- Explicitly declare when a story was generated with the help of AI to ensure greater audience trust.

Integrating AI into data storytelling offers extraordinary opportunities but requires careful management to avoid the risk of incorrect or distorted information. If used consciously, AI can become a powerful ally in data communication, allowing you to build clearer, more accessible, and impactful narratives. Always remember that the value of a data story lies not only in its ability to engage the audience but also in its truthfulness, completeness, and impartiality. Only by ensuring these principles can you harness the potential of AI to create stories that inform and inspire without sacrificing data integrity.

AI is not here to replace you but to amplify your ability to tell compelling stories. But the real power still lies with you. You must guide AI, question its outputs, and ensure the insights you share are meaningful. I have witnessed the transformation AI brings to data storytelling, and I know you can harness it, too. The choice is yours: Will you embrace this revolution or be left behind? The future of data storytelling is unfolding—make sure your voice is part of it.

Takeaways

- AI can be used as a creator, optimizer, reviewer, and assistant, improving the storytelling process without replacing human critical thinking.
- Use AI to tailor stories to target audiences, optimizing language, tone, and format to maximize understanding and engagement.
- Test AI-generated stories by simulating audiences and gathering feedback on the information's clarity, emotional impact, and completeness.

- The use of AI in data storytelling requires attention to avoid issues of toxicity, incompleteness, inaccuracy, and bias.
- The AI revolution in data storytelling is already underway and offers great potential for those who can consciously and strategically integrate it into their creative process.

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